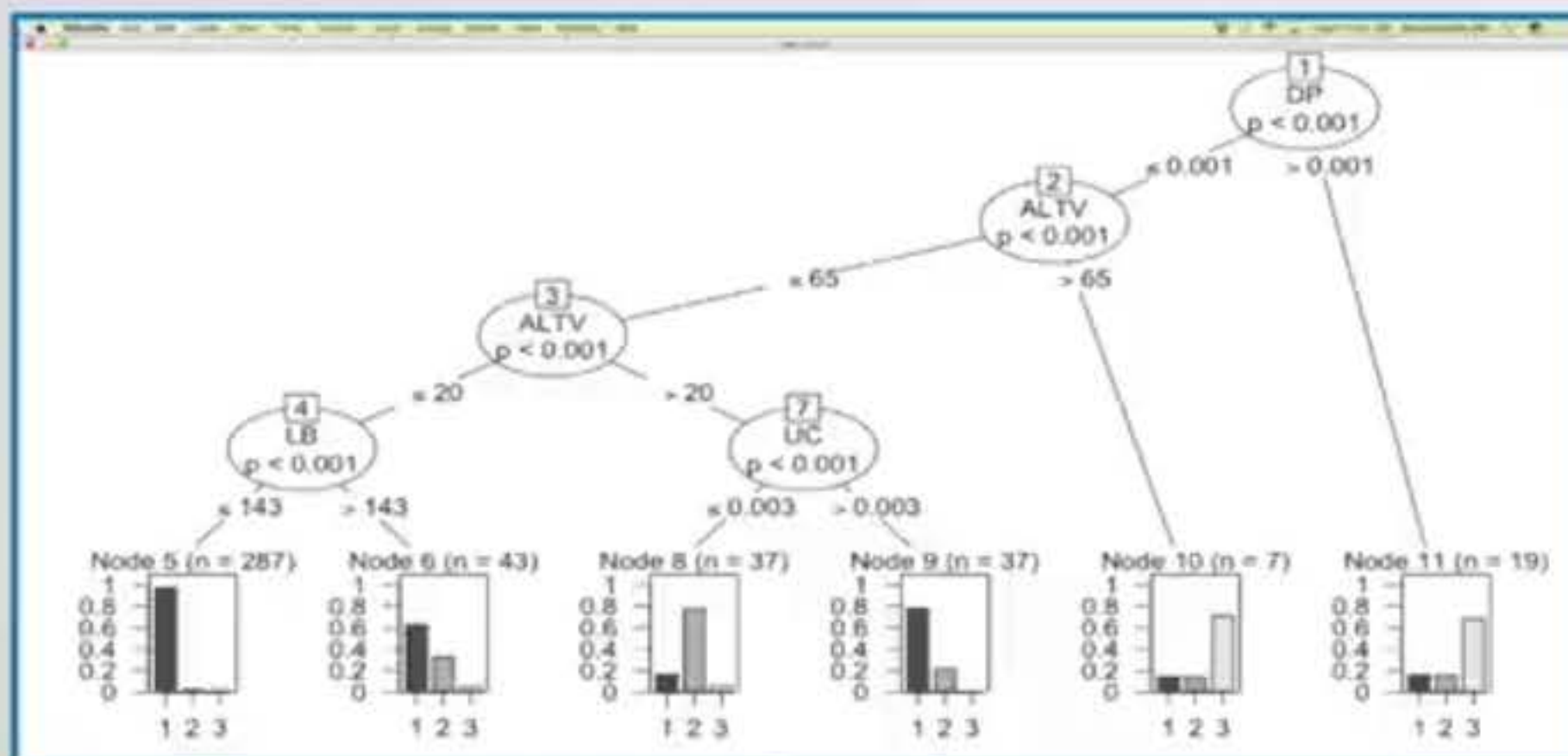
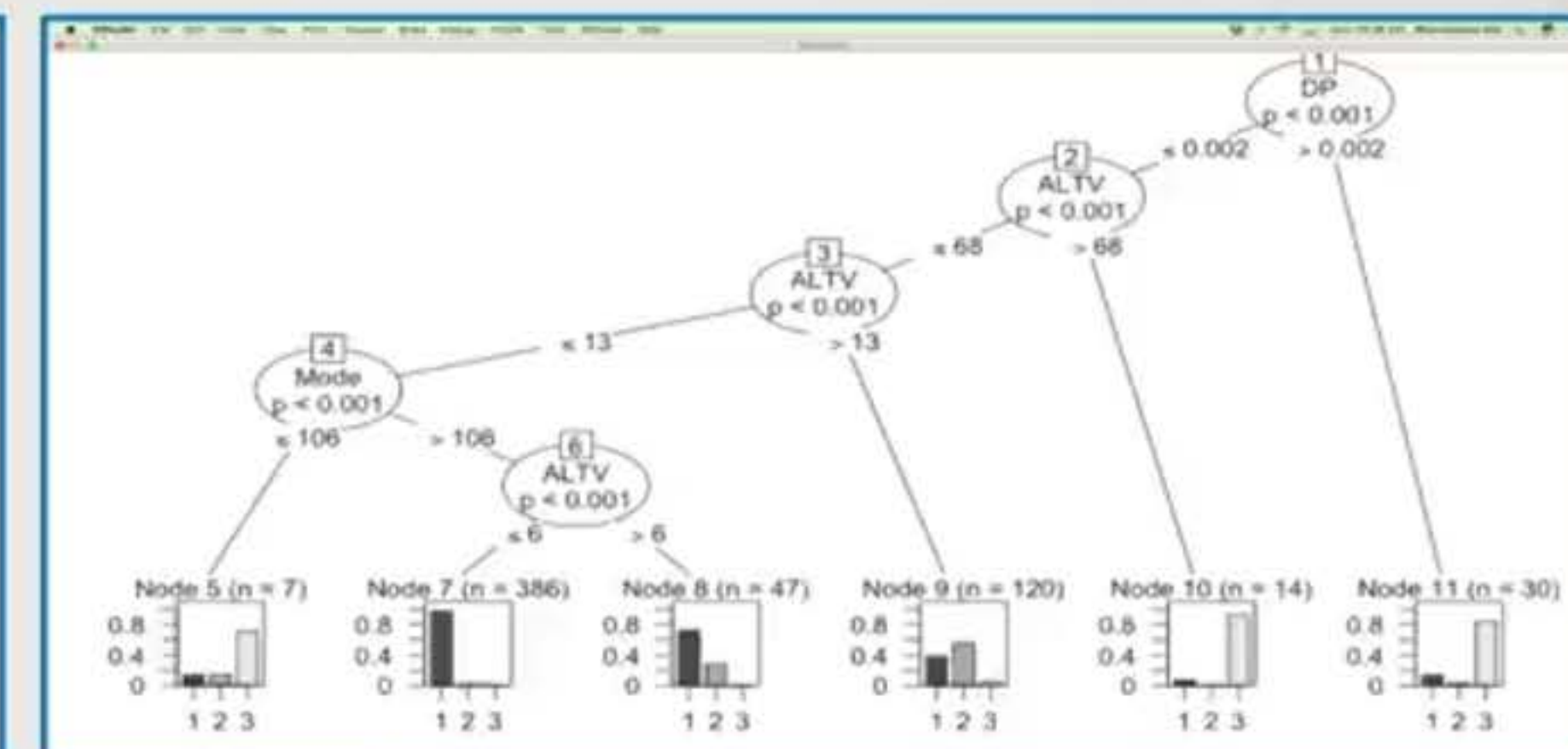
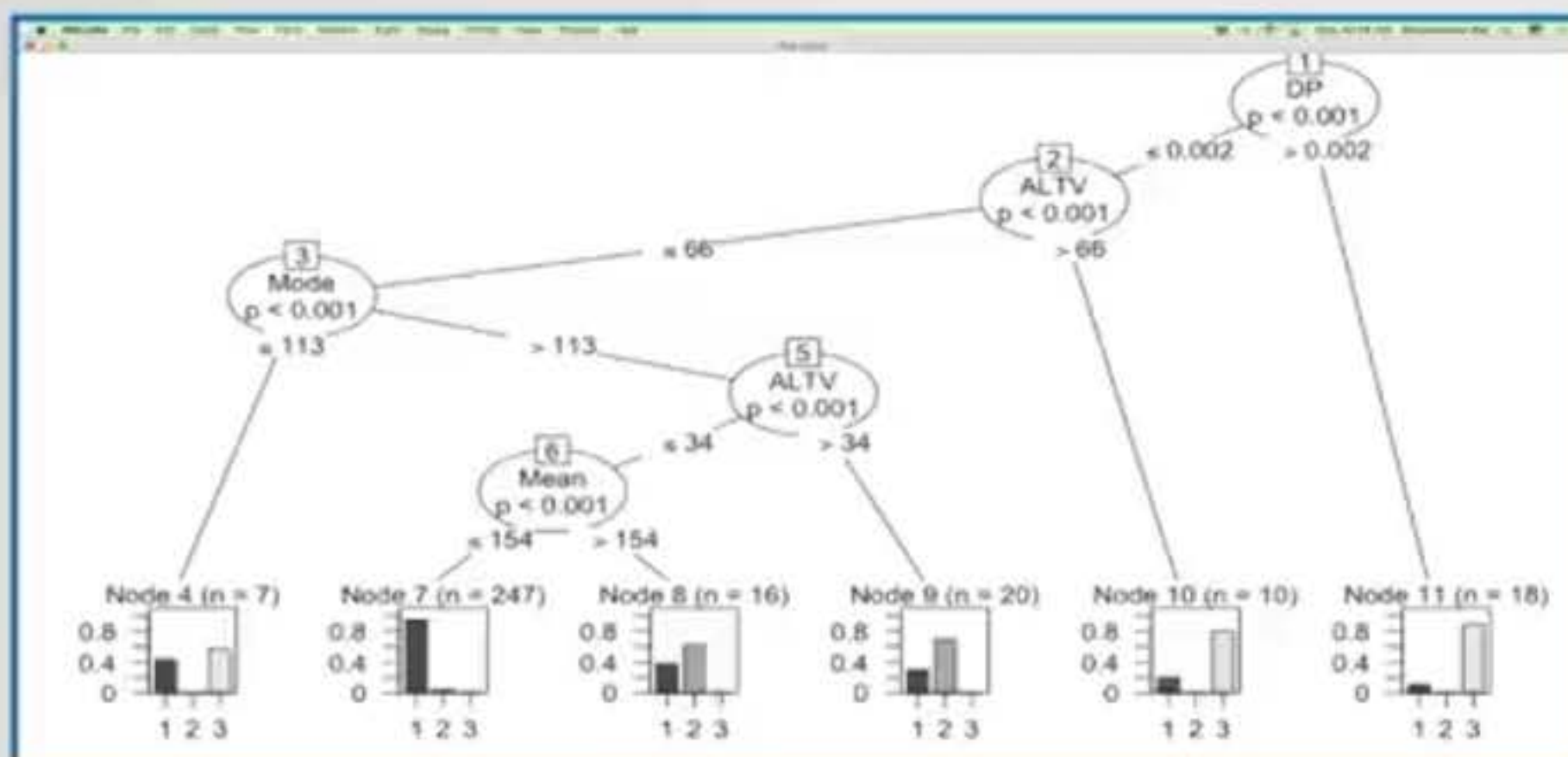
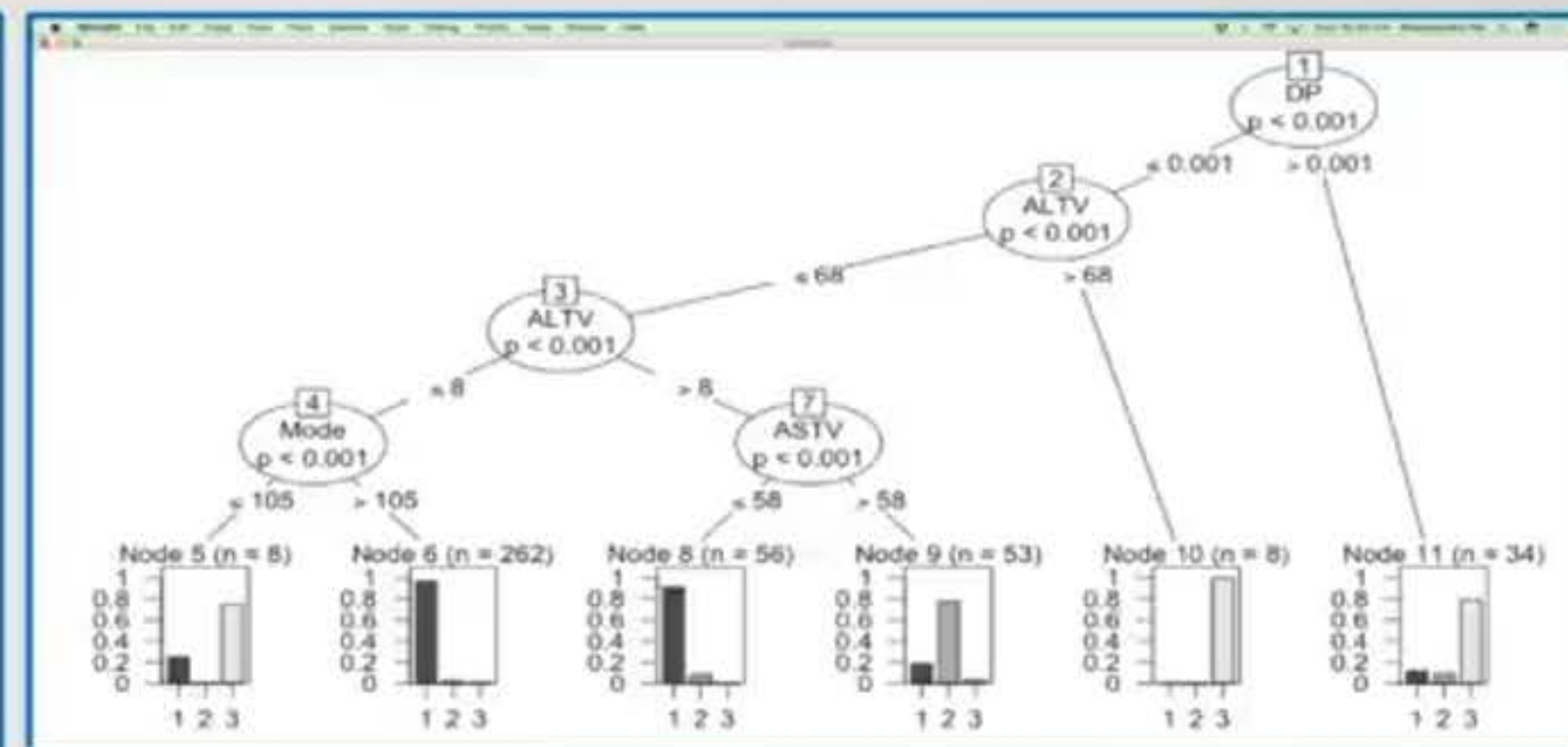
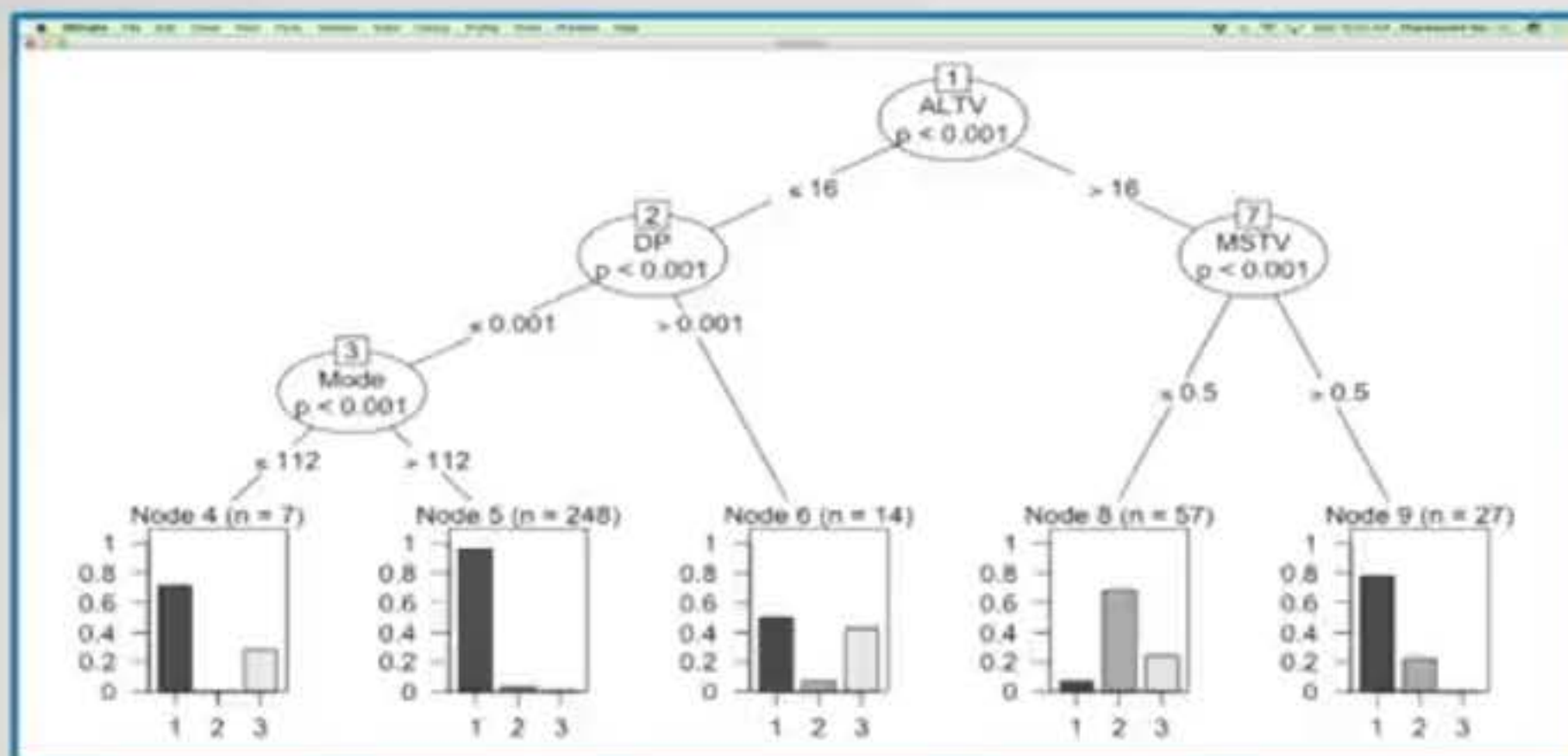


RANDOM FOREST

- Random Forest developed by aggregating trees. ✓
- Can be used for classification or regression. ✓
- Avoids overfitting.
- Can deal with large number of features.
- Helps with feature selection based on importance.
- User-friendly: only 2 free parameters
 - Trees - ntree, default 500
 - Variables randomly sampled as candidates at each split - mtry, default is $\text{sq. root}(p)$ for classification & $p/3$ for regression

STEPS

1. Draw ntree bootstrap samples. ✓
2. For each bootstrap sample, grow un-pruned tree by choosing best split based on a random sample of mtry predictors at each node.
3. Predict new data using majority votes for classification and average for regression based on ntree trees.



New Patient Data

DP = 0.00003

ALT V = 30

MSTV = 0.2

Mode = 10

ASTV = 80

UC = 0.1

OUT OF BAG (OOB) ERROR

- For each bootstrap iteration and related tree, prediction error using data not in bootstrap sample (also called out of bag or OOB data) is estimated.
 - Classification: Accuracy
 - Regression: R-Sq & RMSE

rf

